

Distance sensor

DETECT

DISTANCE

What does it do?

A distance sensor measures how far away something is without touching it.

It works like a bat. One side sends out a click too high for you to hear, the other side listens for the echo, and the time it takes gives the distance.

Your projects can use it to react as something comes closer, or to measure a gap that keeps changing.

Uses



Parking sensor

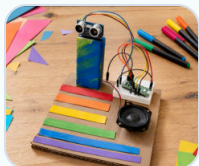
DETECT

Gap to the wall

DECIDE
Is it closer
than 30 cm?

DO

Beep faster



Air harp

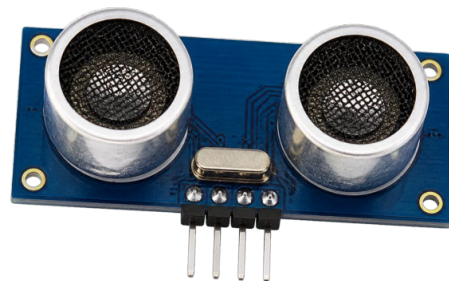
DETECT

How far your
hand is

DECIDE
Which note is
that?

DO

Play the note



VCC

Power,
3V3

Trig

Sends a
click

Echo

Times the
return

GND

Ground

Code

● Raspberry Pi

```
from gpiozero import DistanceSensor

sensor = DistanceSensor(
    echo=18, trigger=17
)

distance = sensor.distance
```

● Raspberry Pi Pico

```
from picozero import DistanceSensor

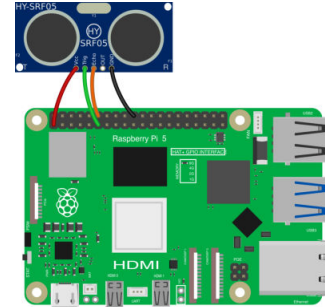
sensor = DistanceSensor(
    echo=15, trigger=13
)

distance = sensor.distance
```

`distance` is in metres and stops at 1. Pass `max_distance=2` to see further, or read `sensor.value` for 0 to 1. Power it from **3V3**, not 5V.

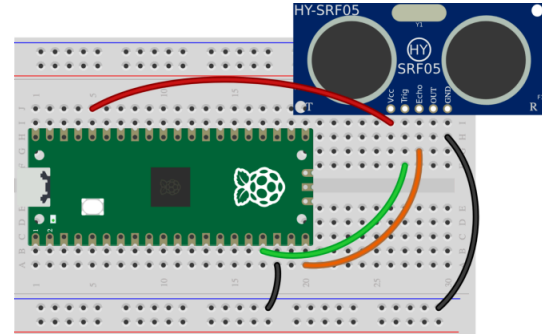
Wiring

RASPBERRY PI 5



Trig **GPIO 17** · Echo **GPIO 18**

PICO



Trig **GP13** · Echo **GP15**